Plan the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use circuit and mechanical evidence to plan a controlled motor-car test.

SUCCESS CRITERIA - I can...

1. Draw the battery-switch-motor circuit.
2. Predict effects of an open switch and a changed connection.
3. Plan one controlled performance comparison.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “circuit” and “motor” to explain your model or result. Sentence starters are optional.

circuit: a connected arrangement of electrical components

motor: a device converting electrical energy to mechanical output

constraint: a requirement or limit that the design must meet

01 Warm-up: bring back an earlier idea

The same flux change through the same coil occurs in 1/3 of the time.
Compare induced-emf magnitude and name a condition for that comparison.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: P = IV; E = PΔt.
Constant power model. Use approved battery, switch and motor. Disconnect before rewiring; keep wheels clear for first power.

Worked reasoning: A classroom car circuit uses a battery, switch and motor in one closed loop. The switch opens the current path; a chassis drawing must also reserve space for safe wiring and moving parts.

Your turn: Draw the circuit separately from a labeled component-placement plan.

Sentence starter: My design shows … . It meets the requirement by … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw and label the battery, switch, motor and wires as a complete loop.
Mark a location where an open connection stops current.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: 1 2 3

04 Apply the physics

During a supplied model-car test, the motor draws 0.50 A at 3 V for 6 s and the car travels 5 m.
Find electrical input power, input energy and average speed. Complete the table and keep these supplied values separate from your team's later measurements.

Sentence starter: I found … [unit]. This value means … in this situation.

Voltage (V)	Current (A)	Time (s)	Distance (m)
3	0.5	6	5

Marzano check - circle: 1 2 3

05 Explain your reasoning

Your team changes wheels, battery and axle alignment together and gets a faster run. Can you attribute the change to wheels alone?
Explain a better comparison.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Electrons in a wire drift at 0.4 mm/s while a lamp responds quickly after a switch closes.
Explain why response time cannot be found by timing one electron across the wire.
Draw the stated circuit and label quantities relevant to the question.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Name two reasons electrical input energy will exceed the car’s final translational kinetic energy.

Sentence starter: I notice … . This supports … because … .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 6 V motor draws 0.8 A normally and 2.4 A when stalled in supplied test data.
Compare input powers and explain a test plan that avoids prolonged stall.

Sentence starter: Both … . They differ in … because … .

Condition	Voltage (V)	Current (A)	Power (W)
Running	6	0.8	____
Stalled	6	2.4	____

Marzano check - circle: 1 2 3